

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering GPA: 4.0 / 4.0</i> Selected Coursework: Large Language Model Systems, Distributed Systems, Machine Learning	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering GPA: 3.72 / 4.0</i> Selected Coursework: Large Language Models in Practice, Operating Systems, Computer Architecture	May 2024

WORK EXPERIENCE

NewsBreak (Venture Team) <i>AI Engineer Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Built agentic search over SSE: deterministic RAG on turn 1, then a bounded tool-calling agent (≤ 3 hops, 5 tools, 45 s) for follow-ups, with client-held chat state (no PII at rest) and server-resolved citations. - Fused keyword, full-text, and bge vector retrieval by RRF, degrading to the surviving legs on failure, and used deterministic entity resolution as the query rewriter, taking real logged queries from 0% to 100% P@1 at 0.4 ms. - Decomposed personalized news alerts into a multi-step LLM pipeline on a Kubernetes CronJob (bge embeddings $\rightarrow$ categorical LLM scoring $\rightarrow$ per-topic percentile gate $\rightarrow$ LLM dedupe $\rightarrow$ push) at ~ 2 min ingest-to-decision and $\sim 1,200$ LLM calls/day; cut filler 34% $\rightarrow$ 4% after 11 days of production shadow. - Root-caused a 65% failure rate in another team’s Redis Streams LLM workers via Langfuse traces as an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways) and a Helm env shadowing the code default. - Built the Python FastAPI / MongoDB backend from its first commit to 89 endpoints with $\sim 1,450$ offline tests and cut a 25–30 request fan-out to one call on a shared cache $78\times$ smaller than per-user keys. - Deployed it on AWS EKS (Docker, Helm, Jenkins, autoscaling 2–4 pods) on day 3, set up Prometheus and Grafana, and fixed a 5-day silent CronJob outage (non-terminal ImagePullBackOff + Forbid concurrency). - Primary backend author for SportsBreak (1.4K App Store ratings); moved nine third-party integrations, including LLM and embedding APIs, server-side so the iOS client ships zero API keys.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
- Led an 8-intern team through Agile sprints to ship a Next.js, FastAPI, and Supabase app on GCP App Engine. - Designed schemas and REST APIs for LangChain LLM resume parsing, suggestions, and AI job matching.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
- Built the LLM mock-interview app from 0$\rightarrow$1 as sole engineer; owned auth, sessions, and the database. - Iterated its AI scoring with recruiters and clients until three 200–500-person companies signed contracts.	

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University <i>Graduate Researcher Advisor: Prof. Yorie Nakahira</i>	February – June 2026
Mapped natural-language instructions to formal safety specs with multi-turn LLMs and implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models).	

SKILLS

AI/ML	Python, LLM tool calling / agents, MCP, RAG, LangChain, PyTorch, CUDA, hybrid retrieval
Backend	Java, FastAPI, REST APIs, SSE, SQL, PostgreSQL, MySQL, MongoDB, Redis Streams
Infra	AWS, GCP Vertex AI, Kubernetes (EKS), Docker, Helm, CI/CD, Prometheus, Grafana, Linux
Languages	C, Go, TypeScript, JavaScript, Swift, Bash
Certifications	AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner

PROJECTS

- my-av:** Built a PyTorch CNN-GRU that predicts steering and speed from 1 s video, with a data pipeline over 33 hours of Comma2k19; trained on local CUDA and GCP Vertex AI and ran streaming inference.
- Wisconsin Autonomous:** Integrated wheel-speed encoders on the competition rover and streamed real-time data to the onboard NVIDIA Jetson.